

CSM-WNet: A Multi-modal Feature Fusion Network for MRI Brain Tumor Segmentation

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Abstract

A critical challenge in 3D multi-modal MRI analysis stems from the inherent information redundancy and coupling across different imaging modalities. This complexity makes it difficult to effectively integrate multi-modal information and extract meaningful features from the intricate modality-specific characteristics. Commonly used but simplistic fusion strategies, such as early fusion, which treats multi-modal data as multi-channel inputs, or late fusion, which aggregates results after independent processing, often fail to exploit the complementary relationships among imaging sequences fully. Meanwhile, brain tumors exhibit high intrinsic heterogeneity with sub-regions including enhancing tumor edema and necrotic core, demonstrating substantial variations in shape, size, location, and textural characteristics, while also possessing frequently ill-defined boundaries against surrounding healthy tissues. To address these challenges we propose CSM-WNet a novel framework designed explicitly for 3D multi-modal MRI segmentation. CSM-WNet reconfigures the traditional skip connections of the U-Net architecture by integrating up-sampling modules thereby enabling effective fusion of multi-scale image features across different resolutions. Furthermore, the introduced CSMix feature fusion mechanism performs refined extraction along both channel-wise and spatial dimensions, allowing the model to learn and leverage distinctive characteristics from various MRI modalities effectively. Experimental results on the BraTS2021 and BraTS2024 datasets demonstrate that CSM-WNet significantly outperforms existing state-of-the-art models in segmenting various tumor sub-regions, while maintaining a light-weight model structure that ensures computational efficiency and fast inference speed.

CCS Concepts

• Computing methodologies → Image segmentation.

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Keywords

Brain Tumor Segmentation, Multi-modal MRI, Convolutional Neural Networks, Feature Fusion

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1 Introduction

Gliomas are among the most prevalent and aggressive primary brain tumors, characterized by their high mortality and recurrence rates [17]. Precise segmentation of tumor sub-regions (e.g., enhancing tumor, peritumoral edema, and necrotic core) from magnetic resonance imaging (MRI) is a critical prerequisite for diagnosis, surgical planning, treatment monitoring, and survival prediction [16]. Magnetic resonance imaging (MRI) provides a non-invasive solution for this task, with different sequences (modalities) offering complementary information: T1-weighted (T1), contrast-enhanced T1-weighted (T1ce), T2-weighted (T2), and Fluid-Attenuated Inversion Recovery (FLAIR) sequences highlight different anatomical and pathological characteristics [14]. Traditionally, this process relies on manual delineation by radiologists, which is not only time-consuming and labor-intensive but also suffers from inter- and intra-observer variability [18]. Considering the multi-modal nature of brain tumor MRI data and the inherent challenges associated with manual segmentation, developing automated methods capable of effectively synergizing complementary information from these modalities is of paramount clinical importance and represents a significant challenge in medical image analysis [28].

However, the clinical deployment of multi-modal MRI segmentation faces unique and substantial hurdles that distinguish it from single-modality tasks. A primary bottleneck is the structural and spatial misalignment between modalities. Unlike single-sequence processing, multi-modal models strictly assume pixel-to-pixel correspondence across T1, T2, and FLAIR sequences. In clinical practice, patient motion during varying acquisition times or geometric distortions in specific sequences (e.g., EPI distortions) inevitably introduce registration errors, which can severely mislead deep learning

models that rely on early channel concatenation [30]. Furthermore, the issue of missing modalities presents a critical barrier to usability. Standard models typically require a complete set of four sequences (T1, T1ce, T2, FLAIR); however, clinical protocols are often incomplete due to time constraints or contrast agent contraindications. A model trained on full modalities often fails catastrophically when a single modality is absent, rendering it unusable in flexible clinical workflows [2]. Finally, distinct modalities possess a semantic gap regarding pathological features—for instance, edema is hyperintense on FLAIR but hypointense on T1, creating conflicting boundary information. Naive fusion strategies often struggle to reconcile these discrepancies, leading to segmentation artifacts at the interface of tumor sub-regions [7].

In response to these challenges, deep convolutional neural networks [13] (CNNs), particularly U-Net and its numerous variants, have become the de facto standard and have demonstrated remarkable success in medical image segmentation [23]. Models like 3D U-Net and V-Net effectively extract spatial features from 3D volumes, capturing crucial contextual information [4, 21]. For multi-modal data, some methods employ late fusion by training separate encoders for each modality and merging features at the decoder stage [10]. While these approaches have achieved promising results, they often exhibit critical limitations in their feature fusion capabilities [31]. Late fusion may struggle to capture the complex, high-level inter-dependencies between modalities early in the feature extraction process [11]. This leads to a suboptimal fusion representation, where the rich, complementary information within the multi-modal data is not fully exploited. Therefore, there is a pressing need for a more sophisticated and effective feature fusion mechanism that can dynamically model the correlations between modalities, selectively emphasize informative features, and suppress redundant or noisy ones, thereby generating a more powerful and discriminative multi-modal representation to boost segmentation accuracy [26].

To address the critical limitation of insufficient utilization of rich complementary information in multi-modal MRI data by current brain tumor segmentation networks, this paper proposes a novel segmentation framework named CSM-WNet. This model introduces innovative modifications to traditional skip connections based on the classical U-Net architecture. Firstly, a Cascaded Up-sampling Module is incorporated to effectively aggregate multi-modal features from different resolutions in the encoder path through a progressive feature fusion strategy, thereby providing richer contextual information for the decoder. Secondly, an innovatively designed CSMix Module is proposed to perform adaptive calibration and refinement of the fused multi-modal features, ensuring that the unique complementary information from modalities such as T1, T1ce, T2, and FLAIR is fully captured and leveraged. These improvements work synergistically to significantly enhance the model's segmentation capability for tumor regions, particularly those with blurred boundaries and irregular shapes, ultimately improving the accuracy and robustness of the segmentation results. The main contributions of this study are as follows:

- By introducing the Cascaded Up-sampling Module, the output channels of encoder modules at different hierarchical

resolutions of the U-Net are interconnected, forming a "W"-shaped network model structure and providing the decoder with more abundant contextual information.

- The design of the CSMix Module enables the full capture and utilization of complementary information from multi-modal data, significantly enhancing the model's segmentation capability for tumor regions.

2 Related Work

Convolutional Neural Networks (CNNs) form the foundation of medical image segmentation, with U-Net established as a landmark architecture for 2D/3D segmentation through its symmetric encoder-decoder with skip-connection [4, 23]; addressing the challenge of manual model design, nnUNet [11] proposed a powerful framework that automatically configures the entire training pipeline, including preprocessing, network topology, and postprocessing, frequently achieving state-of-the-art performance across diverse challenges through robust, dataset-specific customization rather than a fixed architectural innovation.

Integrating Transformer architectures [27] into 3D medical image segmentation has addressed key limitations of CNNs in modeling long-range dependencies. TransUNet [3] pioneered a hybrid architecture by combining a Vision Transformer [6] (ViT) encoder with a CNN-based U-Net decoder, effectively leveraging global context while preserving local spatial precision; however, its extension to 3D volumes remains constrained by high computational complexity and memory demands due to the dense self-attention mechanism. UNETR [9] reformulated 3D segmentation as a sequence-to-sequence problem, utilizing a pure Transformer encoder to capture multi-scale global features and connecting it to a CNN decoder via skip connections. At the same time, it achieves state-of-the-art performance on benchmarks like BTCV and MSD. However, it requires significant computational resources and may struggle with fine-grained details due to the loss of local inductive bias. SwinUNETR [8] incorporated a shifted window-based self-attention mechanism to reduce computational complexity while maintaining global modeling capabilities, enabling efficient processing of 3D volumes; nevertheless, its performance relies heavily on window size configuration and may still face challenges in capturing extremely long-range dependencies across large volumes.

Recent advancements in 3D medical image segmentation have explored channel fusion mechanisms to enhance multi-modal feature integration. UMamba [19], for instance, combines the U-Net architecture with a Mamba-based encoder that leverages state-space models (SSM) for global context modeling; though its effectiveness heavily depends on the quality of cross-modal feature alignment and may struggle with highly heterogeneous modality-specific characteristics. RWKV [22] (Receptance-Weighted Key-Value), known for its breakthrough design in sequence modeling, has been increasingly applied to computer vision tasks. RWKV-UNet [12] leverages the robust temporal and channel-wise properties of RWKV to enhance feature extraction and multi-scale information fusion capabilities, achieving state-of-the-art performance on several 2D medical benchmark datasets such as Synapse, ACDC [24], and BUSI [1]. In the future, combining RWKV with CNN architectures is

expected to become a highly promising research direction in 3D medical image segmentation.

3 Method

3.1 CSM-WNet Framework

The name "U-Net" derives from its distinctive U-shaped symmetric encoder-decoder architecture. This framework progressively extracts contextual information and high-level semantic features of images through the encoder (contracting path), while gradually restoring spatial resolution via the decoder (expanding path) to achieve pixel-level precise localization. The remarkable performance of U-Net and its numerous variants in medical image segmentation fully demonstrates the efficacy of this U-shaped encoder-decoder structure. Therefore, CSM-WNet, adopts the encoder-decoder structure as its foundational backbone.

However, U-Net and its variants demonstrate limited capability in processing multi-modal MRI inputs. Their skip connection structures insufficiently exploit cross-modal feature fusion and extraction, resulting in the model's inability to fully leverage complementary information from different modalities across varying resolutions to delineate tumor boundaries precisely. Thus, to enhance the model's capability in feature extraction and fusion from multi-modal MRI data, we introduce a novel Channel Feature Fusion Module integrated into the skip connections between the encoder and decoder of CSM-WNet. This module is designed to dynamically align and aggregate multi-scale features across different resolutions, enabling more effective capture of heterogeneous tumor characteristics and improving segmentation accuracy. The network architecture of CSM-WNet is illustrated in Fig. 1.

The input to the model, denoted as X , is a 5-dimensional tensor:

$$X \in \mathbb{R}^{B \times C \times H \times W \times D} \quad (1)$$

In this tensor, B denotes the batch size, $C = 4$ represents the number of MRI modalities (T1, T1ce, T2, and Flair), while H, W, D correspond to the height, width, and depth of the volumetric image, respectively. For the input sequence X , CSM-WNet first processes it through an Encoder, which progressively down-samples the spatial resolution while expanding the number of feature channels. This Encoder module comprises a two-layer CNN convolutional block with residual connections. The computation at each layer is defined as follows:

$$X^i = \text{CNR}(\text{CNR}(X^{i-1})) + C'(X^{i-1}) \in \mathbb{R}^{B \times 2^i C \times \frac{H}{2^i} \times \frac{W}{2^i} \times \frac{D}{2^i}} \quad (2)$$

where, CNR denotes a single convolutional layer module, comprising a convolutional operation, normalization, and an activation function. We employ Instance Normalization (IN) as our normalization method. Unlike Batch Normalization, which operates across the batch dimension, IN performs normalization entirely within individual samples, thereby strictly preserving inter-sample independence. This is particularly advantageous for brain tumor data containing multi-modal information, as IN is more effective than BN at retaining the unique characteristic features of each input modality. For the activation function, we adopt LeakyReLU, which serves as a direct and effective improvement over standard ReLU [20]. Introducing a slight negative slope mitigates the 'dying ReLU' problem while maintaining all beneficial properties of ReLU in the

positive region [29] (computational simplicity and mitigation of gradient vanishing). The term C' represents an additional 2×2 convolutional branch applied to the raw input, serving to preserve original visual information and transmit it as residual information to the next layer. Notably, to ensure sufficient retention of input information, the Input Block only expands the channel dimensionality of the input without altering its spatial resolution. Furthermore, to reduce computational overhead, the final block in the Encoder stage does not further expand the channel dimensions but only performs spatial down-sampling.

The encoder in CSM-WNet performs two primary functions: progressively reducing the spatial resolution of the input image and, concurrently, relaying feature maps from each hierarchical level to the corresponding decoder layer through skip-connection. This design enables the decoder to leverage multi-scale contextual information for accurate, step-wise image reconstruction.

3.2 CSMix Module

The CSMix Module serves as the core component of CSM-WNet, designed to enhance the Skip-connection of the U-shaped segmentation network through integrated channel and spatial mix mechanisms. By incorporating Up-sampling layers during the channel mix stage, the module enables cross-level fusion of low-resolution features, breaking the conventional pattern where encoder and decoder features are independently transmitted via skip connections at corresponding resolutions. This innovative approach constructs a unique W-shaped network architecture. Specifically, each CSMix Block receives feature maps from the encoder at the same resolution level and incorporates up-sampled outputs from the lower-level CSMix Block, facilitating multi-scale information propagation.

Subsequently, in the Spatial Mix stage, the CSMix Module performs spatial transformation and reconstruction of the multi-channel fused features, further integrating multi-modal characteristics within the 3D space. A dedicated attention mechanism ensures thorough extraction of fused features, enabling the segmentation model to effectively learn and utilize inter-modal characteristics. This enhanced capability allows the model to handle complex multi-modal medical images containing intricate feature information, improving overall segmentation accuracy in brain tumor tasks. A schematic diagram of the CSMix module is illustrated in Fig. 2.

In the Channel Mix phase, the CSMix module simultaneously accepts input features from the encoder at the current stage and the output features from the preceding CSMix module. Specifically, the features from the preceding CSMix first undergo an up-sampling operation to align their dimensions with those of the current encoder, facilitating feature aggregation via element-wise addition. Subsequently, the aggregated features bifurcate into two branches, denoted as X and X' . For branch X , the features are directly flattened into a 1D vector without Channel Shift. Conversely, branch X' is subjected to a Channel Shift operation prior to 1D flattening. This procedure is formally expressed as follows:

$$X = \text{Flat}(\text{Ups}(X_{\text{CSMix}}) + X_{\text{Encoder}}) \in \mathbb{R}^{B, C, L}, L = (H \times W \times D) \quad (3)$$

$$X' = \text{Flat}(\text{Shift}(\text{Ups}(X_{\text{CSMix}}) + X_{\text{Encoder}})) \in \mathbb{R}^{B, C, L}, L = (H \times W \times D) \quad (4)$$

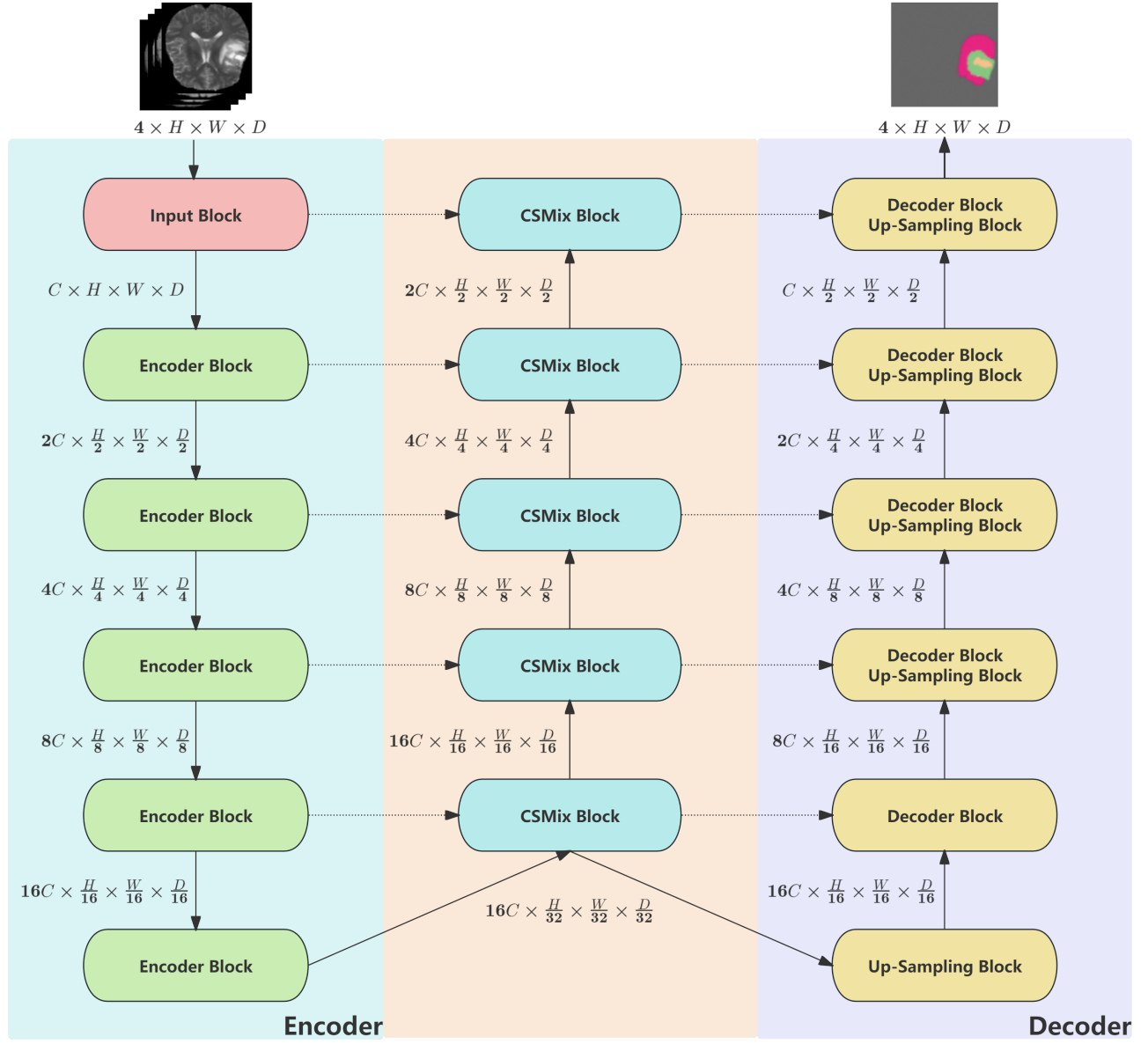


Figure 1: The overall structure of CSM-WNet.

where, L denotes the dimension after flattening the height, width and depth of the 3D image into a single dimension. Ups denotes the upsampling layer, $Shift$ represents the application of the Channel Shift algorithm, and $Flat$ refers to the 1D flattening operation. $X_{Encoder}$ and X_{CSMix} denote the features output from the hierarchical encoder and the preceding CSMix module, respectively, with data dimensions given by:

$$X_{Encoder} \in \mathbb{R}^{B \times 2^i C \times \frac{H}{2^i} \times \frac{W}{2^i} \times \frac{D}{2^i}} \quad (5)$$

$$X_{CSMix} \in \mathbb{R}^{B \times 2^{i+1} C \times \frac{H}{2^{i+1}} \times \frac{W}{2^{i+1}} \times \frac{D}{2^{i+1}}} \quad (6)$$

where, i represents the stage depth of the current CSMix module, where i ranges from 1 to 4.

The principle of Channel Shift is illustrated in Fig. 3. This operation applies 3D spatial translations across eight distinct directional combinations: (up, forward, left), (up, forward, right), (up, backward, left), (up, backward, right), (down, forward, left), (down, forward, right), (down, backward, left), and (down, backward, right). By partitioning the image channels into different translation branches, it performs pixel-wise shifts within each channel branch. The Channel Shift mechanism enhances the differentiation of fused features

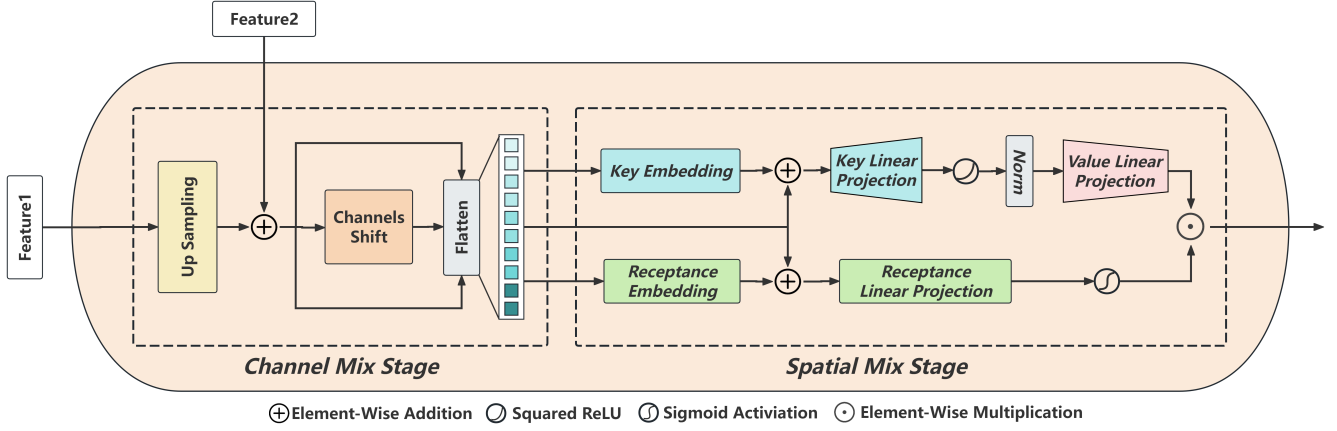


Figure 2: Diagram of the CSMix Module.

by introducing controlled spatial variations, thereby preventing excessive homogenization of multi-modal feature information during repeated fusion operations. This process ensures the diversity, richness, and robustness of the model's multi-modal feature extraction capabilities.

During the Spatial Mix stage, the model first generates semantic embeddings for both Key and Receptance from the original input features that have not undergone Channel Shift processing. These embeddings are combined with the Channel Shift processed feature inputs via Element-Wise Addition to achieve feature fusion. The corresponding formulation is presented below:

$$X_{Key} = X \cdot Embd_{Key} + X' \cdot (1 - Embd_{Key}) \quad (7)$$

$$X_{Rcep} = X \cdot Embd_{Rcep} + X' \cdot (1 - Embd_{Rcep}) \quad (8)$$

The X_{Key} branch first undergoes a linear projection to expand its semantic dimensionality, followed by weight activation and normalization. It is then reduced back to a lower dimension via another linear projection to obtain the value. This value is subsequently subjected to Element-Wise Multiplication with the X_{Rcep} branch, which has been processed by a separate linear projection and an activation function, yielding the final extracted features. The corresponding formulation is presented below:

$$Output = LP_{Value}(Norm(Act_{Key}(LP_{Key}(X_{Key})))) \cdot Act_{Rcep}(LP_{Rcep}(X_{Rcep})) \quad (9)$$

where, LP denotes the linear projection layer applied to the feature tensor. Act_{Key} and Act_{Rcep} represent the Squared ReLU and Sigmoid activation, respectively. $Norm$ refers to Layer Normalization, which is particularly well-suited for processing flattened 1D data.

3.3 Loss Fuction

During the training process, considering the multi-class nature of the brain tumor segmentation task, we adopt a combined loss function comprising the multi-class Dice Loss [25] and a region-based Focal Loss [15] for model optimization. The mathematical

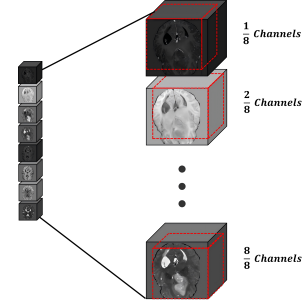


Figure 3: Diagram of the Channel Shift Principle. The Channel Shift operation performs pixel-wise displacements on 3D voxel clusters along eight distinct spatial directions.

expression for the Focal Loss component is defined as follows:

$$L_{Focal} = - \sum_{i=1}^N \alpha (1 - p_i)^\gamma y_i \log(p_i) \quad (10)$$

where, N denotes the number of semantic classes in the segmentation task. p_i represents the predicted probability map, and y_i corresponds to the ground-truth mask. The term α denotes the class-balancing weight, while γ is the focusing parameter that reduces the loss contribution from well-classified samples. Focal Loss provides pixel-level fine-grained gradients, ensuring the model does not overlook challenging pixels—such as those near tumor boundaries or within small lesions—often critical to achieving high segmentation accuracy.

To simultaneously ensure global optimization of the segmented regions, we incorporate a multi-class weighted Dice Loss. This loss function maximizes the overlap area between the predicted and ground-truth regions, thereby promoting better overall shape integrity and topological consistency in the segmentation results,

while avoiding fragmented or disjointed predictions. The mathematical formulation of the Dice Loss is defined as follows:

$$L_{Dice} = 1 - \frac{2 \times (\sum_{i=1}^N p_i y_i + \mu)}{\sum_{i=1}^N p_i + \sum_{i=1}^N y_i + \mu} \quad (11)$$

where, μ is a smoothing factor, set to $1e^{-7}$ in our experiments to ensure numerical stability and training stability. The overall loss function is a weighted linear combination of the Focal Loss and Dice Loss. While the Dice Loss ensures global integrity of the segmented regions at a macroscopic level, the Focal Loss enhances the model's discriminative capability for hard-to-classify pixels at a microscopic level. The combined loss function aims to jointly improve the model's final segmentation accuracy and robustness through this synergistic collaboration. The mathematical formulation is as follows:

$$L_{total} = \lambda \cdot L_{Focal} + (1 - \lambda) \cdot L_{Dice} \quad (12)$$

where, the coefficient λ serves as a hyper-parameter designed to balance the contributions of both loss components to the total optimization objective. For our experimental setup, its value was empirically fixed at 0.5.

Table 1: Data Number on BraTS2021 and BraTS2024 Datasets.

Dataset	Training Sets	Validation Sets	Testing Sets
BraTS2021	1000	125	126
BraTS2024	1080	135	136

4 Results

4.1 Datasets

We utilized the BraTS2021 and BraTS2024 brain tumor MRI datasets released by MICCAI in 2021 and 2024, respectively [5]. Both datasets consist of volumetric images with dimensions of $155 \times 240 \times 240$. Each case includes four MRI modalities (T1, T1ce, T2, and FLAIR) and is annotated with four labels: 0 for background, 1 for edema, 2 for necrosis, and 4 for enhancing tumor. The segmentation targets were defined as the enhancing tumor (ET, label 4), tumor core (TC, labels 1+4), and whole tumor (WT, labels 1+2+4). The datasets were partitioned into training, validation, and test sets in an 8:1:1 ratio. Specifically, BraTS2021 contains 1,000 cases for training, 125 for validation, and 126 for testing, while BraTS2024 includes 1,080 training cases, 135 validation cases, and 136 test cases, illustrated in Table 1.

To mitigate issues such as excessive noise and data imbalance caused by large black backgrounds, the original image dimensions of $155 \times 240 \times 240$ were cropped to $128 \times 128 \times 128$. This cropping strategy aims to maximize the visibility of lesion regions while preserving essential information. Subsequently, the ground truth masks underwent Z-score standardization, and the four original sub-labels within the MRI data were re-indexed. The reassigned labels are as follows: 0 for background, 1 for necrosis, 2 for edema, and 3 for enhancing tumor. This re-indexing redefines the segmentation targets as enhancing tumor (ET, label 3), tumor core (TC, labels 2+3), and whole tumor (WT, labels 1+2+3), thereby simplifying the model training process.

4.2 Metrics and Implementation Details

Our network model and training framework were implemented on a system running Ubuntu 20.04, with CUDA version 11.8 and PyTorch version 2.0.1. The model was trained on a single A800 GPU with a batch size of 4. We employed AdamW as the optimizer, with the learning rate set to 0.0005. In our experiments, the Dice Score, Hausdorff Distance 95 percent (HD95), Mean Intersection over Union (mIoU) and Recall were used as evaluation metrics to assess the performance of all models compared.

The Dice Score (Dice Similarity Coefficient, DSC) quantifies the similarity between two sample sets. In the context of segmentation tasks, it measures the degree of overlap between the model's predicted segmentation and the ground truth annotation. The Mean Intersection over Union (mIoU) computes the intersection ratio to the union of the predicted and ground truth regions. Recall (Sensitivity or True Positive Rate) evaluates the model's ability to identify all positive instances. In medical applications, high recall is of paramount clinical importance, as it corresponds to a lower risk of missed diagnoses. The corresponding mathematical formulations are given below:

$$Dice = \frac{2 \times TP}{2 \times TP + FP + FN} \quad (13)$$

$$MIoU = \frac{1}{n} \sum_{i=1}^N \frac{TP}{TP + FP + FN} \quad (14)$$

$$Recall = \frac{TP}{TP + FN} \quad (15)$$

where, TP denotes the number of pixels correctly predicted as positive examples by the model. FP represents the number of pixels erroneously predicted as positive examples. FN indicates the number of pixels incorrectly predicted as negative examples. N refers to the total number of classes in the segmented image.

The Hausdorff Distance 95 percent (HD95) is a critically important evaluation metric in medical image segmentation. It is primarily employed to measure the contour alignment accuracy between segmentation results and ground truth labels, particularly focusing on the worst-case alignment error. The formal definition of the HD95 metric is as follows:

$$HD95 = \text{Max}(h_{95}(A, B), h_{95}(B, A)) \quad (16)$$

where, $h(A, B) = \text{Max}_{a \in A} \text{Min}_{b \in B} \|a - b\|$. A, B denote sets of pixels, where a, b represent individual pixel points within these sets. h_{95} denotes that the value at the 95 percent of all calculated distances is adopted as the final result.

4.3 Comparative Experiments

We conducted a comprehensive comparative analysis between the proposed CSM-WNet and several state-of-the-art (SOTA) methods, including U-Net, U-Net++, V-Net, UNETR, SwinUNETR, nnUNet, AttentionUNet, TransUNet, and UMamba. To ensure a fair comparison, all methods were trained under identical experimental settings, utilizing the same loss function, learning rate warm-up strategy, and early stopping mechanism. The maximum number of training epochs was set to 300 for all models, with early stopping being triggered after 10 consecutive epochs without improvement. All reported evaluation metric scores represent the average values

Table 2: Comparison with The SOTA Methods on BRATS2021 and BraTS2024 Datasets.

Model	#Params(M)	Dice Score(%) $\uparrow$				Hausdorff95 $\downarrow$			
		ET	TC	WT	Avg.	ET	TC	WT	Avg.
U-Net	10.24	78.72/75.24	77.81/75.08	75.15/74.36	77.23/74.89	4.619/5.535	9.121/9.805	11.50/11.98	8.412/9.106
U-Net++	4.890	80.94/79.73	80.55/79.44	78.36/79.04	79.95/79.4	4.487/5.134	9.021/9.654	10.86/11.17	8.124/8.653
V-Net	8.762	50.31/57.42	59.35/65.11	74.73/78.04	61.46/66.86	7.649/7.133	9.937/9.852	11.43/12.03	9.671/9.673
UNETR	97.60	69.13/68.61	64.31/65.46	51.00/49.05	61.48/61.04	6.331/6.389	10.25/10.47	15.20/16.33	10.59/11.06
SwinUNETR	14.03	78.92/77.91	78.07/76.94	74.75/73.24	77.25/76.03	4.901/5.196	9.337/9.697	11.52/11.97	8.588/8.954
nnUNet	30.45	80.59/79.35	81.03/80.69	77.64/75.86	79.75/78.64	4.124/5.633	8.912/9.988	11.31/12.34	8.114/9.322
AttentionUNet	5.679	81.83/78.72	83.93/79.24	84.49/74.82	83.41/77.59	4.467/5.437	8.800/9.814	10.80/11.71	8.023/8.989
TransUNet	126.3	83.25/81.61	85.54/84.33	88.62/85.62	85.8/83.85	4.522/4.981	8.763/9.332	10.78/10.77	8.020/8.361
UMamba(Bot)	39.03	84.02/80.91	86.17/82.54	87.83/82.84	86.01/82.1	3.895/4.963	8.366/9.187	10.72/11.25	7.659/8.467
UMamba(Enc)	40.86	85.02/83.54	87.61/85.75	88.88/85.87	87.17/85.05	3.760 /5.105	7.830/8.860	10.36/11.38	7.318/8.448
Ours	55.96	86.02/84.86	88.87/87.41	89.87/87.45	88.25/86.57	4.007/4.705	7.798/8.368	9.808/10.82	7.204/7.966

obtained from a five-fold cross-validation process, with numerical results retained to four decimal places.

As shown in the Table 2, on the BraTS2021 dataset, CSM-WNet achieved Dice Scores of 86.02, 88.87, 89.87, and 88.25 for Enhancing Tumor, Tumor Core, Whole Tumor, and their average, respectively. The corresponding Hausdorff95 scores were 4.007, 7.798, 9.80, and 9.204. On the BraTS2024 dataset, the Dice scores were 84.86, 87.41, 87.45, and 86.57, with Hausdorff95 scores of 4.705, 8.368, 10.82, and 7.966 for the respective regions. CSM-WNet demonstrated superior performance compared to existing methods in most tumor sub-region evaluations for both BraTS2021 and BraTS2024 datasets, as reflected in both Dice and Hausdorff95 metrics. Fig. 4 presents a 2D visual comparison between our proposed segmentation model and the aforementioned SOTA models in the brain tumor segmentation task. The qualitative results presented in the figures clearly illustrate CSM-WNet’s enhanced precision in segmenting tumor areas and its improved capability in maintaining the structural integrity of tumor contours.

Furthermore, CSM-WNet achieved marginally lower Hausdorff95 scores in segmenting certain tumor sub-regions than state-of-the-art models such as TransUNet and UNETR. However, it exhibits a significantly smaller parameter count and a more streamlined architecture. This efficiency advantage indicates that CSM-WNet can efficiently segment 3D multi-modal medical image lesions with substantially reduced computational overhead, further validating the effectiveness of its improvements in the multi-modal feature fusion strategy.

Table 3: Ablation Study of CSMix Module and W-shape on BraTS2021 DATASET.

CSMix	W-shape	Dice Score(%) $\uparrow$			
		ET	TC	WT	Avg.
		82.12	83.38	84.83	83.44
	✓	82.97	85.13	86.86	84.99
✓		83.88	85.69	86.90	85.49
✓	✓	86.02	88.87	89.87	88.25

4.4 Ablation Study

To assess the impact of different parameters and modules on CSM-WNet’s feature fusion and tumor segmentation performance, we conducted a series of ablation studies focusing on two key aspects: the CSMix module and the W-shaped feature up-sampling pathway.

As shown in Table 3, the experimental results on the BraTS2021 dataset demonstrate that incorporating the W-shaped network architecture into the backbone network achieved Dice score improvements of 0.81, 1.75, 2.03, and 1.55 for Enhancing Tumor, Tumor Core, Whole Tumor, and their average, respectively. The introduction of the CSMix module, which enables channel-wise feature fusion through cascaded up-sampling, yielded improvements of 1.76, 2.31, 2.07 and 2.05 over the backbone. When both components were integrated, CSM-WNet achieved even greater performance gains in Dice scores. These results strongly validate the superior capability of the CSMix module and the W-shaped feature fusion architecture in processing multi-modal medical MRI data with complex information. The proposed components effectively enhance the model’s ability to extract and leverage complementary information across different modalities, leading to more efficient and accurate tumor segmentation.

Furthermore, we investigated the impact of spatial transformation diversity enabled by a key component of the CSMix module: the Channel Shift operation. During feature extraction, the Channel Shift component performs pixel-wise shifting along eight distinct directional combinations within the three-dimensional space, enhancing feature differentiation across channel dimensions. To examine the correlation between this differentiation capability and the functional design of Channel Shift, we reduced the number of shifting directions to four—limiting transformations to the height and width dimensions while preserving the depth dimension unchanged. This modification simulates the feature extraction process typically applied to 2D image slices, allowing us to validate the generalizability and extensibility of the Channel Shift mechanism in 2D image segmentation tasks.

Based on Table 4, the 3D Channel Shift component demonstrates notable improvements in Dice scores for Enhancing Tumor, Tumor Core, Whole Tumor, and their average compared to its 2D counterpart, with increases of 0.15, 0.94, 0.53 and 0.54, respectively. These findings confirm that by performing differentiated processing of

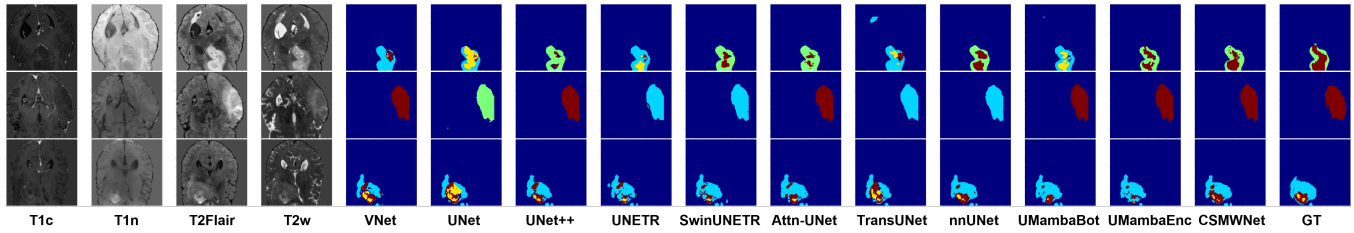


Figure 4: Examples of segmentation results from multiple methods. From left to right: T1c, T1n, T2Flair, T2w modality input image, segmentation results from various comparison algorithms, the proposed CSM-WNet, and Ground Truth (GT).

multi-modal features across a broader range of spatial directions, the 3D Channel Shift component offers more significant advantages in enhancing the richness of multi-modal feature representations. Furthermore, this result validates the extensibility of the Channel Shift mechanism in image segmentation tasks, indicating its broad application potential in both 3D and 2D medical image segmentation scenarios.

Table 4: Ablation Study of Channel Shift on BraTS2024 DATASET.

Channel Shift	Dice Score(%) $\uparrow$			
	ET	TC	WT	Avg.
$\times$	84.26	86.74	86.54	85.85
2D	84.71	86.45	86.92	86.03
3D	84.86	87.41	87.45	86.57

5 Conclusion

Three-dimensional brain tumor MRI segmentation plays a pivotal role in clinical decision-making by precisely delineating tumor boundaries in 3D space. To address the multi-modal characteristics of MRI, we propose CSM-WNet, incorporating a novel W-shaped network architecture. This design enhances skip-connections via up-sampling modules, enabling comprehensive fusion of MRI features across different resolutions. Furthermore, the introduced CSMix feature fusion mechanism performs refined extraction along both channel-wise and spatial dimensions, effectively leveraging distinctive characteristics from various modalities. Consequently, segmentation accuracy for tumor subregions is significantly improved. Experimental results on the BraTS2021 and BraTS2024 datasets demonstrate that CSM-WNet significantly outperforms state-of-the-art models while maintaining a lightweight structure for computational efficiency.

However, it is worth noting that our evaluations were primarily conducted on standardized public datasets. As such, the model’s generalization capability across multi-center clinical data with heterogeneous acquisition protocols and varying image quality remains to be fully verified. Future research will focus on validating the robustness of our strategy in real-world clinical scenarios and exploring its application to additional downstream tasks.

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